

Principal Component Regression (PCR)

Collinearity, Regularization View, Model Selection, and Diagnostics
Question Set with Solutions

TTK4260: Multivariate Data Analysis and ML

A. Motivation and Core Idea

Question 1 (*Why do we need PCR? Collinearity*)

[4 points]

Consider a multiple linear regression (MLR) model $y = \mathbf{X}\boldsymbol{\theta} + \varepsilon$ with mean-centered columns in $\mathbf{X}$. Explain (i) what it means that the predictors are *collinear*, and (ii) why collinearity can make the ordinary least squares (OLS) estimate $\hat{\boldsymbol{\theta}}_{\text{OLS}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ numerically unstable and have high variance.

Solution. (i) Collinearity means that some columns of $\mathbf{X}$ are close to linear combinations of others, so $\mathbf{X}$ has directions in predictor space with (near) zero variance and $\mathbf{X}^\top \mathbf{X}$ is (nearly) singular.

(ii) In OLS we must invert $\mathbf{X}^\top \mathbf{X}$. If $\mathbf{X}^\top \mathbf{X}$ is ill-conditioned (very small eigenvalues), then small perturbations in the data (noise, rounding, sampling variation) cause large changes in $(\mathbf{X}^\top \mathbf{X})^{-1}$, inflating the variance of $\hat{\boldsymbol{\theta}}_{\text{OLS}}$ and producing unstable coefficients (large magnitude, sign flips) even when predictions may not change much.

Question 2 (*PCR pipeline*)

[4 points]

Write the three main steps of Principal Component Regression (PCR) for a single response variable y . Assume $\mathbf{X}$ is mean-centered (and possibly scaled). Use PCA notation $\mathbf{X} = \mathbf{T}\mathbf{P}^\top$ and let $\mathbf{T}_r$ denote the first r score columns.

Solution. A standard PCR procedure is:

- (1) **PCA on $\mathbf{X}$:** Compute loadings $\mathbf{P}$ and scores $\mathbf{T} = \mathbf{X}\mathbf{P}$ (equivalently $\mathbf{X} = \mathbf{T}\mathbf{P}^\top$).
- (2) **Truncate:** Keep only the first r principal components: $\mathbf{T}_r = \mathbf{X}\mathbf{P}_r$.
- (3) **Regress y on scores:** Fit OLS in score space,

$$\hat{\boldsymbol{\beta}} = (\mathbf{T}_r^\top \mathbf{T}_r)^{-1} \mathbf{T}_r^\top \mathbf{y}, \quad \hat{\mathbf{y}} = \mathbf{T}_r \hat{\boldsymbol{\beta}}.$$

Optionally back-project to original variables: $\hat{\boldsymbol{\theta}}_{\text{PCR}} = \mathbf{P}_r \hat{\boldsymbol{\beta}}$ so that $\hat{\mathbf{y}} = \mathbf{X} \hat{\boldsymbol{\theta}}_{\text{PCR}}$.

Question 3 (*PCR vs OLS in matrix form*)

[6 points]

$$\hat{\mathbf{y}}_{\text{PCR}} = \mathbf{X}\mathbf{P}_r(\mathbf{T}_r^\top \mathbf{T}_r)^{-1} \mathbf{T}_r^\top \mathbf{y}.$$

Compare this with the OLS projection matrix $\mathbf{H}_{\text{OLS}} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ and explain the *key difference*.

Solution. Because $\hat{\mathbf{y}}_{\text{PCR}} = \mathbf{T}_r \hat{\boldsymbol{\beta}}$ and $\mathbf{T}_r = \mathbf{X}\mathbf{P}_r$, we have

$$\hat{\mathbf{y}}_{\text{PCR}} = \mathbf{T}_r(\mathbf{T}_r^\top \mathbf{T}_r)^{-1} \mathbf{T}_r^\top \mathbf{y} = \mathbf{X}\mathbf{P}_r(\mathbf{T}_r^\top \mathbf{T}_r)^{-1} \mathbf{T}_r^\top \mathbf{y}.$$

OLS instead uses

$$\hat{\mathbf{y}}_{\text{OLS}} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

Key difference: OLS inverts $\mathbf{X}^\top \mathbf{X}$ directly (problematic if ill-conditioned). PCR first projects $\mathbf{X}$ to an r -dimensional orthogonal score space and inverts $\mathbf{T}_r^\top \mathbf{T}_r$, which is diagonal (or well-conditioned) because the score vectors are mutually orthogonal. PCR also restricts the fitted model to the subspace spanned by the first r PCs.

Question 4 (*Why each β_j is a simple scalar*)

[5 points] Let $\mathbf{t}_j$ be the j -th score vector (column of $\mathbf{T}$) and suppose the scores are orthogonal. Show that the OLS coefficient for regressing y on the first r scores satisfies

$$\hat{\beta}_j = \frac{\mathbf{t}_j^\top \mathbf{y}}{\mathbf{t}_j^\top \mathbf{t}_j}, \quad j = 1, \dots, r.$$

Interpret this geometrically.

Solution. With orthogonal columns, $\mathbf{T}_r^\top \mathbf{T}_r$ is diagonal with entries $\mathbf{t}_j^\top \mathbf{t}_j$. Thus

$$\hat{\boldsymbol{\beta}} = (\mathbf{T}_r^\top \mathbf{T}_r)^{-1} \mathbf{T}_r^\top \mathbf{y}$$

reduces componentwise to $\hat{\beta}_j = (\mathbf{t}_j^\top \mathbf{y}) / (\mathbf{t}_j^\top \mathbf{t}_j)$.

Geometry: $\mathbf{t}_j^\top \mathbf{y} / \|\mathbf{t}_j\|^2$ is the scalar that projects $\mathbf{y}$ onto the direction $\mathbf{t}_j$; PCR is projecting $\mathbf{y}$ onto the span of the selected score vectors.

Question 5 (*Back-projection to original variables*)

[5 points] Suppose $\mathbf{X}$ is mean-centered and $\mathbf{T}_r = \mathbf{X} \mathbf{P}_r$. Show that the regression coefficients in the original variable space are

$$\hat{\boldsymbol{\theta}}_{\text{PCR}} = \mathbf{P}_r \hat{\boldsymbol{\beta}}.$$

If y was also mean-centered, how do you write the final prediction model for an uncentered new sample $\mathbf{x}_{\text{new}}$?

Solution. From $\hat{\mathbf{y}} = \mathbf{T}_r \hat{\boldsymbol{\beta}} = (\mathbf{X} \mathbf{P}_r) \hat{\boldsymbol{\beta}} = \mathbf{X} (\mathbf{P}_r \hat{\boldsymbol{\beta}})$, we identify $\hat{\boldsymbol{\theta}}_{\text{PCR}} = \mathbf{P}_r \hat{\boldsymbol{\beta}}$.

If $\mathbf{X}$ was centered with mean $\bar{\mathbf{x}}$ and y with mean $\bar{y}$, then for a new sample $\mathbf{x}_{\text{new}}$ the centered predictor is $\tilde{\mathbf{x}}_{\text{new}} = \mathbf{x}_{\text{new}} - \bar{\mathbf{x}}$, and

$$\hat{y}_{\text{new}} = \bar{y} + \tilde{\mathbf{x}}_{\text{new}}^\top \hat{\boldsymbol{\theta}}_{\text{PCR}}.$$

(If scaling was used, apply the same scaling to $\tilde{\mathbf{x}}_{\text{new}}$ before prediction.)

B. Geometry and Regularization View

Question 6 (*Project, then regress*)

[4 points] Describe the geometric interpretation of PCR in predictor space. Why can PCR reduce variance compared to OLS?

Solution. Geometrically, PCR first projects $\mathbf{X}$ onto the r -dimensional subspace spanned by the first r loading vectors (high-variance directions in $\mathbf{X}$), forming scores $\mathbf{T}_r$. Then it performs regression in that reduced orthogonal coordinate system.

Variance is reduced because low-variance (often noise-dominated) directions of $\mathbf{X}$ —which correspond to small eigenvalues of $\mathbf{X}^\top \mathbf{X}$ and cause large OLS coefficient variance—are discarded. This trades some bias (due to truncation) for lower variance.

Question 7 (*PCR as shrinkage / hard thresholding*)

[6 points] Assume $\mathbf{X}$ has SVD $\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^\top$

with singular values $s_1 \geq \dots \geq s_p$ and right singular vectors $\mathbf{v}_j$. Write the OLS solution in this basis and explain how PCR modifies it when keeping r components.

Solution. In the SVD basis (for mean-centered $\mathbf{X}$), the OLS estimator can be written as

$$\hat{\boldsymbol{\theta}}_{\text{OLS}} = \sum_{j=1}^p \frac{\mathbf{u}_j^\top \mathbf{y}}{s_j} \mathbf{v}_j \quad (\text{equivalently } \hat{\boldsymbol{\theta}}_{\text{OLS}} = \mathbf{V}\mathbf{S}^{-1}\mathbf{U}^\top \mathbf{y} \text{ when full rank}).$$

PCR keeps only the first r directions:

$$\hat{\boldsymbol{\theta}}_{\text{PCR}} = \sum_{j=1}^r \frac{\mathbf{u}_j^\top \mathbf{y}}{s_j} \mathbf{v}_j.$$

Thus PCR applies a *hard threshold* shrinkage: coefficients along components $j > r$ are set to zero (discarded), while the kept components are unchanged relative to the OLS components in those directions.

Question 8 (*Bias-variance tradeoff*)

[5 points] Explain qualitatively how the number of retained PCs r affects bias and variance in PCR. What happens at the extremes $r = 0$ and $r = p$ (full rank)?

Solution. As r increases, PCR includes more directions of $\mathbf{X}$, typically decreasing bias (more flexible model) but increasing variance (more sensitivity to noise, especially along low-variance directions). At $r = 0$, the model predicts only the mean of y (after centering, $\hat{\mathbf{y}} = \mathbf{0}$; uncentered, $\hat{y} = \bar{y}$), giving high bias but minimal variance. At $r = p$ (and $\mathbf{X}$ full rank), PCR becomes equivalent to OLS, giving minimum bias among linear unbiased estimators but potentially high variance under collinearity.

Question 9 (*Head-to-head: PCR vs ridge*)

[6 points] PCR is often compared to ridge regression. State one similarity and one key difference between PCR and ridge regression in terms of shrinkage in the principal-component basis.

Solution. Similarity: Both methods stabilize estimation under collinearity by shrinking the solution in the principal-component directions of $\mathbf{X}$, reducing variance from small-eigenvalue directions.

Difference: PCR uses *binary* shrinkage factors ($d_j \in \{0, 1\}$): it keeps the first r directions unchanged and discards the rest. Ridge uses *continuous* shrinkage factors $d_j = \frac{s_j^2}{s_j^2 + \lambda}$ (in the SVD basis), so all components are retained but shrunk, with stronger shrinkage for small s_j .

C. Model Selection and Cross-Validation

Question 10 (*Selecting r by cross-validation*)

[6 points] Describe a practical procedure to choose the number of components r using K -fold cross-validation (CV). What performance metric would you use for a regression problem?

Solution. A typical procedure:

- (1) Split the data into K folds.

- (2) For each candidate $r \in \{1, \dots, r_{\max}\}$ and each fold k :
 - (a) Fit PCA on the *training* $\mathbf{X}$ only; obtain $\mathbf{P}_r$.
 - (b) Compute training scores $\mathbf{T}_{r,\text{train}} = \mathbf{X}_{\text{train}}\mathbf{P}_r$ and fit $\hat{\boldsymbol{\beta}}$ by OLS.
 - (c) Project the validation predictors using the *training* loadings: $\mathbf{T}_{r,\text{val}} = \mathbf{X}_{\text{val}}\mathbf{P}_r$ and predict $\hat{\mathbf{y}}_{\text{val}} = \mathbf{T}_{r,\text{val}}\hat{\boldsymbol{\beta}}$.
- (3) Aggregate the validation error across folds for each r and select the r with lowest error (or using a one-standard-error rule).

A common metric is RMSE (root mean squared error) or MSE on validation folds; R^2 on held-out data can also be reported.

Question 11 (*Avoiding data leakage*)

[5 points] In PCR with CV, why is it incorrect to compute PCA loadings $\mathbf{P}$ once on the full dataset and reuse them inside each fold? What is the correct approach?

Solution. Computing $\mathbf{P}$ on the full dataset uses information from the validation fold (its predictor distribution) during training, which is *data leakage*. This typically makes CV error overly optimistic because the projection basis was influenced by the held-out samples.

Correct approach: within each CV split, fit PCA (and all preprocessing such as centering/scaling) using only the training fold, then project the validation fold with the *training* PCA model.

Question 12 (*Explained variance vs predictive power*)

[6 points] You find that the first two PCs explain 90% of the variance in $\mathbf{X}$, but using only $r = 2$ components gives poor prediction of y . Give two reasons why high explained variance in $\mathbf{X}$ does not guarantee good predictive performance for y .

Solution. Two common reasons:

- (1) PCA is unsupervised: PCs maximize variance in $\mathbf{X}$, not covariance with y . The directions that best predict y may have low variance in $\mathbf{X}$ and appear in later PCs.
- (2) The relationship between $\mathbf{X}$ and y may align with a direction that is *not* among the dominant variance directions (e.g., a subtle but informative contrast between variables), so truncating early may discard predictive signal.

Additionally, if noise dominates early variance directions (e.g., strong nuisance variation), the leading PCs can be predictive of nuisance rather than y .

D. Diagnostics and Interpretation

Question 13 (*Predicted vs observed*)

[4 points] You have a fitted PCR model and compute predictions on a test set. What patterns in a predicted-vs-observed plot indicate (i) a good model, (ii) systematic bias, and (iii) heteroscedasticity or model mismatch?

Solution. (i) Good model: points lie close to the 45° line with roughly constant scatter across the range.

(ii) Systematic bias: curvature or consistent deviation from the line (e.g., underpredicting large values and overpredicting small values).

(iii) Heteroscedasticity or mismatch: scatter increases with the magnitude of y , or distinct fan-shaped patterns; clusters/stripes can indicate missing categorical structure or nonlinear effects not captured by the linear model.

Question 14 (*Residuals vs fitted*)

[5 points] What do you look for in a residuals-vs-fitted plot for a PCR model? Name two warning signs and what they might imply.

Solution. Ideally residuals are centered around zero with no pattern and roughly constant variance. Warning signs include:

- (1) **Curvature/pattern:** suggests nonlinearity or missing interactions; linear PCR may be insufficient.
- (2) **Increasing spread (fan shape):** suggests heteroscedasticity; consider variance-stabilizing transforms, weighted regression, or different modeling.

Outlying residuals may indicate influential points or model breakdown for particular operating regimes.

Question 15 (*Coefficient paths*)

[6 points] Define what is meant by a *PCR coefficient path* as a function of the number of retained components r . How can coefficient paths help interpret collinearity and model stability?

Solution. A coefficient path plots each original-variable coefficient $\hat{\theta}_j(r)$ versus r . It shows how estimated effects change as more PCs are included.

Under strong collinearity, OLS-like solutions (large r) can produce unstable coefficients (large swings, sign changes), while moderate r often yields smoother, more stable coefficients. Thus coefficient paths help identify a region where coefficients stabilize (suggesting a sensible r) and reveal variables whose apparent influence is sensitive to including low-variance components.

Question 16 (*Interpreting PCR in an application*)

[6 points] In a process setting (e.g., many correlated sensors $\rightarrow$ product quality y), you fit PCR with a small r and achieve good prediction. Give two ways PCR can be used for *insight* beyond prediction.

Solution. Two common uses:

- (1) **Latent factor interpretation:** The retained PCs can represent underlying process modes (e.g., “temperature level”, “flow balance”). Examining loadings and scores can reveal which sensors move together and which operating regimes correspond to quality changes.
- (2) **Actionable variable effects:** Back-projected coefficients $\hat{\theta}_{\text{PCR}}$ (or their paths vs r) can indicate which sensors/inputs are most associated with quality, guiding feature selection, control priorities, or further experiments.

E. Limitations and Pitfalls

Question 17 (*When PCR can fail*)

[6 points] Give an example scenario (conceptual, not necessarily numeric) where PCR can perform poorly even though it reduces collinearity problems. What alternative method is often introduced to address this limitation?

Solution. PCR can fail when the direction in $\mathbf{X}$ that is most predictive of y has *low variance* in $\mathbf{X}$ and appears in later PCs. If we choose r based on explained variance in $\mathbf{X}$ (or too small via CV noise), we may discard the predictive direction and get poor performance.

A common alternative is **Partial Least Squares (PLS)**, which constructs latent variables to maximize covariance between $\mathbf{X}$ and y (supervised dimension reduction).

Question 18 (*Centering and scaling choices*)

[5 points] In PCR, why is mean-centering of $\mathbf{X}$ almost always required? When is scaling (standardization) important, and what can go wrong if you skip it?

Solution. Mean-centering ensures the first PC captures directions of variation around the mean and avoids the mean vector dominating the decomposition; it also aligns PCR with a model that includes an intercept separately.

Scaling is important when variables are in different units or have very different variances; otherwise PCA (and thus PCR) is dominated by high-variance variables, potentially ignoring informative but smaller-scale predictors. Skipping scaling can lead to components that reflect measurement units rather than underlying structure and can harm predictive performance and interpretability.

Question 19 (*Reporting a “full PCR story”*)

[6 points] List four items you would include when reporting PCR results in a scientific/engineering setting to tell the “full story” (not just a single RMSE number).

Solution. A solid report often includes:

- (1) **Preprocessing details:** centering/scaling and how applied to train/test.
- (2) **Model selection:** CV scheme, chosen r , and uncertainty (e.g., one-standard-error rule).
- (3) **Predictive diagnostics:** predicted vs observed (CV/test), residuals vs fitted, and summary metrics (RMSE, R^2).
- (4) **Interpretation:** explained variance in $\mathbf{X}$, coefficient paths, and/or discussion of loadings/scores for retained PCs.

Optionally include sensitivity checks, outlier/influence assessment, and comparison to baselines (OLS, ridge, PLS).